

**Final Year
Internship Report**

On

Early Detection of Autism Spectrum Disorder

In partial fulfilment of the requirement for the PG degree of
MSc – Big Data Analytics

Submitted By

24-PBD-004 – Vedant R Chaklasiya
(Semester – IV MSc BDA)

Under the supervision of

Dr. Pravida Raja



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CERTIFICATE

This is to certify that

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of Semester-IV of MSc – Big Data Analytics,
of the Department of Data Sciences has successfully completed
the internship and worked on the project titled
“Early Detection of Autism Spectrum Disorder”
as part of the curriculum of Semester IV of
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Pravids
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**Infividhya****10th April, 2026**

To,
Pravida Raja,
Assistant Professor & HOD,
Department of Data Science
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Dear Ma'am,

This is to inform you that Mr. Vedant Chaklasiya , who is pursuing his M.Sc. in Big Data Analytics from St. Xavier's College, Ahmedabad , has commenced his internship at Infividhya Academy Pvt. Ltd. on 8th December 2025 . He has completed 4 months of internship, and his internship is scheduled to conclude on 8th May 2026 .

If you require any documentation or references, please feel free to reach out to info@infividhya.com.

Thank you for your efforts, and we hope our paths cross again.

Regards,

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Declaration

I, Vedant Chaklasiya, student of M.Sc. Big Data Analytics at St. Xavier's College, Ahmedabad, hereby declare that this internship report titled "Early Detection of Autism Spectrum Disorder" is a record of original work carried out by me during my internship period at InfiVidhya Academy Pvt. Ltd. under the supervision of Dr. Archana Singh and Dr. Sejal Bhavasar.

I further declare that the work contained in the report is original and has been done by me under the general supervision of my supervisors. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this University or any other University of India or abroad.

I have followed the guidelines provided by the department in writing the report. Wherever I have used contents or documentation from other resources, I have mentioned them in the references.

Vedant Chaklasiya

24-PBD-004

M.Sc. Big Data Analytics

St. Xavier's College, Ahmedabad

Acknowledgement

The successful completion of this project at InfiVidhya Academy Pvt. Ltd. would not have been possible without the guidance and support of many individuals. A project of this nature is always the result of the collective efforts of several people, and I feel incredibly fortunate to have received such valuable assistance throughout this journey.

First and foremost, I would like to express my sincere gratitude to Dr. Archana Singh and Dr. Sejal Bhavsar for providing me with the opportunity to undertake this project and for their constant encouragement and support as Co-founders.

This project is the result of consistent effort and dedication. However, it would not have been possible without the continuous support and guidance of my mentors and team members. I feel privileged to have worked with such knowledgeable and supportive individuals for providing me this valuable opportunity to learn and grow in this domain.

A very special and heartfelt thanks to Fr. Dr. David Roy, Director of St. Xavier's College, whose constant encouragement, support, and motivation have always inspired me.

I would like to offer special thanks to Dr. Pravida Raja AC, Head of the Department of Data Science, whose guidance and support played a crucial role in the successful completion of this project. Her constant knowledge sharing and encouragement helped me understand the subject in depth and achieve the desired outcomes.

Finally, I extend my sincere thanks to my teachers, classmates, family, and friends for their unwavering support and encouragement at every stage of this project.

With sincere regards,

Vedant Chaklasiya

24-PBD-004

Executive Summary

This executive summary encapsulates the comprehensive internship experience undertaken at InfiVidhya Academy Pvt. Ltd., fulfilling the practical requirements for the M.Sc. in Big Data Analytics program at St. Xavier's College (Autonomous), Ahmedabad. The internship served as a vital bridge between the theoretical foundations of data science and practical, enterprise-grade technological execution. Working under the guidance of industry experts, the experience provided profound exposure to advanced data engineering, machine learning paradigms, and the rigorous analytical methodologies required to extract actionable insights from complex real-world datasets.

Throughout the tenure, a strong emphasis was placed on mastering a modern data science technology stack. The primary conceptual focus revolved around bioinformatics, predictive modeling, and high-dimensional data processing. Technical proficiencies were heavily developed using Python and its premier scientific libraries, including pandas and NumPy for structural data manipulation, and scikit-learn for algorithmic deployment and statistical evaluation. Furthermore, the internship necessitated leveraging cloud-based computing infrastructure, specifically Google Colab, to handle the immense memory and processing demands of advanced feature engineering and hyperparameter optimization workflows.

The culmination of this internship was the successful execution of a specialized research project titled "Early Detection of Autism Spectrum Disorder". The project aimed to develop an objective, data-driven diagnostic framework utilizing machine learning to analyze transcriptomic profiles from peripheral blood samples, shifting the diagnostic focus from subjective behavioral assessments to underlying molecular biomarkers. The analytical pipeline involved extracting massive, dual-platform genomic microarray datasets and executing complex preprocessing techniques. This included harmonizing structural data disparities, translating hardware-specific proprietary probes into universal Gene Symbols, and aggressively reducing dimensionality through statistical feature selection methodologies like ANOVA.

Following the data harmonization phase, the project utilized automated screening libraries to rapidly prototype multiple algorithmic architectures without human bias. The most viable models including tree-based ensembles, margin-based classifiers, and regularized linear models were then subjected to deep optimization loops utilizing

stratified cross-validation. Ultimately, the project successfully demonstrated that end-to-end machine learning pipelines can effectively isolate sparse, predictive genetic signatures from inherently noisy biological data. This comprehensive internship not only solidified core analytical competencies but also highlighted the profound real-world utility of big data analytics in developing advanced clinical decision-support systems.

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Chapter 1

1.1 About the Organization

InfiVidhya Private Limited is a formally registered, MSME-certified IT services and technology consulting firm operating in Ahmedabad, Gujarat. Co-founded by experienced leaders Dr. Archana Singh and Dr. Sejal Bhavsar, the organization bridges the gap between technological innovation and practical business execution. Positioning itself as a robust technology partner, InfiVidhya specializes in delivering enterprise-grade digital solutions to drive long-term innovation and sustainable business growth.

While maintaining a strong foundation in academic upskilling, the firm's core operations are structured around six key service pillars designed to enhance digital transformation, security, and operational efficiency:

Custom Software & Full-Stack Engineering: Delivering end-to-end product development—from concept to launch—by building scalable web, mobile, and cloud applications using modern, API-driven architectures and high-performance UI stacks like React and Tailwind.

Technology Consulting & Digital Transformation: Partnering with enterprises to align tech strategies with business goals through system audits, cloud adoption, digital process re-engineering, and technical debt mitigation.

Data Analytics, BI & Research Services: Empowering strategic decision-making through advanced data modeling, automated reporting, and predictive analytics utilizing tools like Power BI.

Cybersecurity & Compliance: Safeguarding digital assets via continuous monitoring, rigorous penetration testing, and ensuring strict compliance with major industry standards like ISO, SOC, and GDPR.

IT Talent & Team Augmentation: Accelerating project delivery through flexible engagement models that deploy pre-vetted developers and tech specialists to seamlessly integrate with client workflows.

Corporate Training & Upskilling: Providing customized, hands-on learning paths to keep client teams at the cutting edge of DevOps, cloud, and full-stack technologies.

By synthesizing advanced software engineering, actionable data intelligence, and robust cybersecurity frameworks, InfiVidhya positions itself as a holistic digital partner dedicated to enabling long-term innovation and sustainable business growth.

1.2 About the Department/Course

The M.Sc. in Big Data Analytics program, offered under the Department of Computer Science at St. Xavier's College (Autonomous), Ahmedabad, focuses on the growing importance of data in today's technology-driven world. The program is designed to build a strong foundation in data analysis, statistical modelling, and data-driven decision-making, enabling students to extract meaningful insights from complex datasets.

The course equips students with the ability to analyse historical data, interpret current trends, and make informed predictions about future outcomes. It emphasizes both theoretical understanding and practical implementation, ensuring that students are well-prepared to tackle real-world challenges in the field of data science and analytics. The curriculum also promotes analytical thinking, logical reasoning, and problem-solving skills, which are essential for making effective business and technical decisions.

The program includes a comprehensive blend of core subjects covering various aspects of data science and analytics. These include Statistical Methods, Probability and Stochastic Processes, Linear Algebra, Database Management Systems, Data Mining, Python Programming, Foundations of Data Science, Machine Learning, Data Visualization, Time Series Analysis, Econometrics, and other enabling technologies.

Overall, the M.Sc. in Big Data Analytics program provides a well-rounded academic experience by integrating theoretical knowledge with hands-on learning, thereby preparing students for successful careers in data analytics, data science, and related domains.

1.3 About the Project

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by challenges in social communication, alongside restricted and repetitive

behaviors. Globally, the prevalence of ASD has been steadily rising, with current World Health Organization estimates suggesting that approximately 1 in 100 children are on the autism spectrum. Despite this increasing prevalence, traditional diagnostic methodologies still rely heavily on subjective behavioral observations and psychometric evaluations. This often leads to significant diagnostic delays, sometimes pushing clinical confirmation into later childhood. Early detection is absolutely critical; identifying ASD during the highly plastic, foundational stages of infant neurodevelopment allows clinicians to initiate timely, targeted therapeutic interventions that can profoundly improve a child's long-term cognitive, social, and communicative outcomes.

To address the inherent limitations and delays of behavioral diagnosis, this project explores the powerful application of Machine Learning (ML) to establish an objective, data-driven diagnostic framework. Specifically, the research focuses on a transcriptomic approach, utilizing high-throughput genomic microarray data to analyze intricate gene expression profiles in individuals with ASD compared to neurotypical controls. Machine learning possesses the unique computational ability to process these massive biological datasets, uncovering subtle, multi-gene epistatic interactions that are virtually impossible to detect through standard univariate statistical methods. By shifting the diagnostic focus to molecular biomarkers, the project aims to identify the underlying genetic signatures of ASD long before behavioral symptoms become definitive.

However, applying ML to genomic data presents the significant computational challenge of high dimensionality, where biological features exponentially outnumber patient samples, risking severe algorithmic overfitting. This project systematically tackles this bottleneck by implementing rigorous statistical feature selection methodologies, specifically Analysis of Variance (ANOVA), to isolate a concentrated panel of highly predictive genes. By training advanced classifiers - such as Support Vector Classifiers and Random Forest on this optimized genomic subset, the project successfully constructs a highly sensitive predictive model. Ultimately, this research demonstrates the viability of utilizing machine learning to decode complex transcriptomic data, paving the way for advanced clinical decision support tools and precision medicine in neurodevelopmental care.

Chapter 2

2.1 Statement of Problem

The challenge of diagnosing Autism Spectrum Disorder (ASD) and its associated conditions - such as Asperger's Disorder and Pervasive Developmental Disorder-Not Otherwise Specified (PDD-NOS) - presents a profound clinical dilemma. Traditionally, the diagnostic process relies almost entirely on behavioral observations, psychological evaluations, and the tracking of developmental milestones. While these clinical assessments are the gold standard, they are inherently subjective and deeply dependent on the expertise of the evaluating clinician. Furthermore, behavioral symptoms often overlap with other developmental delays, leading to frequent misdiagnoses or delayed diagnoses. Because early intervention is critically linked to better long-term developmental outcomes, there is an urgent clinical need for an objective, biologically based diagnostic marker that can identify ASD pathophysiology from a standard physiological sample, such as peripheral blood.

Simultaneously, attempting to find this biological marker introduces a severe computational challenge. Genomic datasets, particularly microarray gene expression data, are notoriously complex. The specific dataset targeted for this research (GSE18123) compounds this complexity by being a high-dimensional, multi-platform collection. It initially contains over 87,000 raw genetic features across 285 patient samples. Crucially, these samples were processed on two entirely different hardware platforms (the older GPL570 and the newer GPL6244). This hardware discrepancy creates a massive structural collision, resulting in a dataset riddled with missing values (NaNs) where specific genetic probes exist on one chip but not the other.

Beyond structural issues, the dataset is plagued by severe demographic and clinical skews. There is a heavy class imbalance between neurotypical controls and the highly fragmented on-spectrum diagnostic subgroups. Additionally, the dataset reflects a 4:1 male-to-female ratio, mimicking historical clinical rates but creating a strong algorithmic bias. The core problem this thesis addresses is the design of a sophisticated computational pipeline capable of mathematically harmonizing these disparate hardware outputs, overcoming the "curse of dimensionality," and correcting demographic imbalances to extract a reliable, generalizable biological signature for ASD.

2.2 System Study

A thorough review of the existing literature reveals a distinct gap between traditional psychiatric diagnosis and modern bioinformatics. Historically, the diagnosis of ASD is strictly codified by criteria outlined in manuals like the DSM-5. While clinical consensus is strong, the search for blood-based transcriptomic biomarkers is a relatively novel frontier. Previous studies that have attempted to use machine learning for ASD detection based on gene expression have frequently relied on singular, uniform datasets. They tend to avoid the immense preprocessing complexity associated with merging multi-platform data, thereby severely limiting the amount of available training data and reducing the statistical power of their predictive models.

Furthermore, a significant methodological gap exists in how algorithms are selected in contemporary bioinformatics research. Many existing systems and published papers rely on arbitrary algorithmic selection - researchers manually choose a popular model, such as a Support Vector Machine or a Random Forest, and force the data to fit that architecture. There is a glaring need for automated, unbiased algorithmic screening systems that can test dozens of mathematical architectures simultaneously to empirically prove which model natively suits the genomic data structure.

The feasibility of addressing these gaps is highly promising given modern computational tools. By utilizing programmatic feature engineering - such as mapping proprietary microarray probes to universal Gene Symbols and utilizing intersection strategies - the technical barrier of multi-platform integration can be reliably bypassed. Furthermore, the integration of automated screening libraries (such as LazyPredict) allows for a rapid, unbiased evaluation of model viability. This project fills the existing literature gaps by demonstrating that an end-to-end, automated, and heavily regularized machine learning architecture can successfully untangle dual-platform transcriptomic data to yield a clinically significant diagnostic baseline.

2.3 System Analysis

To successfully decode the GSE18123 dataset, the proposed predictive system must adhere to a strict set of functional and non-functional requirements. Functionally, the system must be capable of programmatically downloading, unpacking, and parsing

complex .soft.gz files directly from the NCBI Gene Expression Omnibus. It must contain logic to identify and map conflicting hardware probe vocabularies (from both GPL570 and GPL6244) into unified biological Gene Symbols. The system must also execute complex data transformations, including aggregating duplicate genes via mean calculations, intersecting datasets, and applying independent standard scaling to eliminate hardware batch effects. Finally, the system requires a functional module to seamlessly transform the highly fragmented, four-class clinical diagnosis target into a simplified, clinically actionable binary framework (1 = Autism Spectrum, 0 = Control).

From a non-functional perspective, the system demands extraordinary computational resources. Processing a transposed matrix of 285 samples against 87,000+ continuous float variables requires highly optimized memory management. Operations such as the ANOVA feature selection (SelectKBest) and hyperparameter Grid Search cross-validation are heavily CPU-intensive and prone to failure on standard consumer-grade hardware. The system also requires a strictly reproducible and stable software environment to ensure library dependencies (such as pandas and scikit-learn) do not conflict during the mathematical modeling phases.

To satisfy these immense computational and environmental requirements, the entire analytical system was designed, coded, and executed exclusively within the Google Colab cloud environment. Utilizing Google Colab was a vital architectural decision. It provided immediate, on-demand access to high-RAM instances and robust cloud processors capable of executing the dense matrix multiplications required for model training. By leveraging this cloud-based infrastructure, the project completely bypassed local hardware bottlenecks, ensuring the uninterrupted execution of the exhaustive LazyPredict screening and the complex, multi-fold cross-validation tuning grids.

2.4 System Design

The architectural blueprint of the predictive system follows a highly structured, sequential logic. This design ensures that raw, chaotic biological signals are systematically filtered, harmonized, and refined into precise mathematical predictions without the risk of data leakage.

Phase 1 - Extraction and Parsing: The pipeline initiates by querying the NCBI GEO database. The system downloads the raw microarray matrices and the corresponding phenotypic metadata. The data is parsed into memory, immediately exposing the structural disconnect between the GPL570 and GPL6244 platforms.

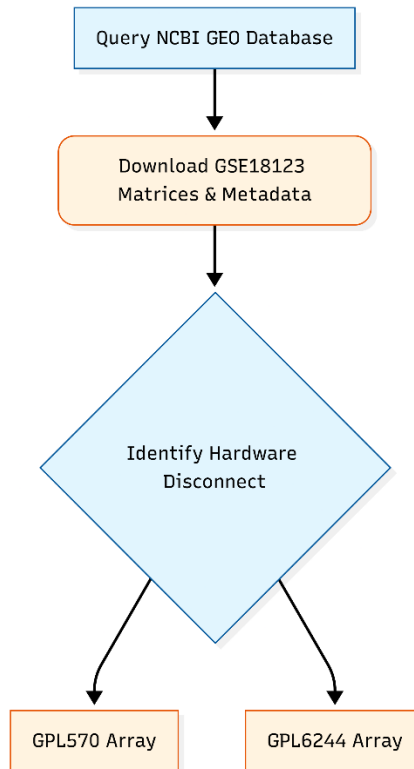


Figure 1: Phase 1 – Extraction and Parsing

Phase 2 - Cleaning and Harmonization: This is the most complex structural node in the architecture. The raw expression values are transposed so that patient samples become rows and genetic probes become columns. The system then utilizes metadata dictionaries to translate hardware probes into universal Gene Symbols. Duplicate genes are smoothed using mean aggregation. The two platforms are mathematically intersected to find common genes, effectively dropping unshared features and resolving all NaN collisions. Finally, the system applies an independent `StandardScaler()` to each platform to neutralize hardware biases before concatenating them into a master harmonized matrix.

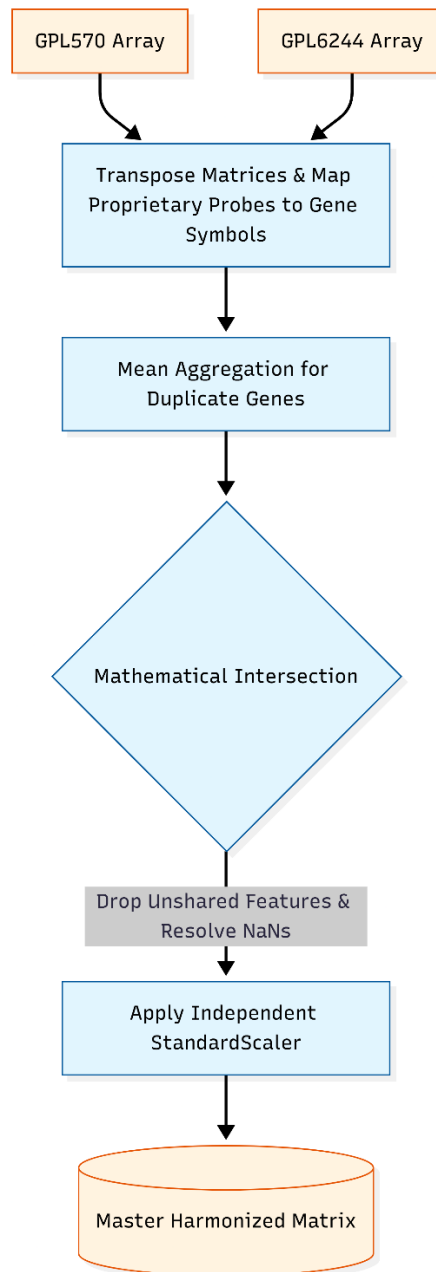


Figure 2: Phase 2 - Cleaning and Harmonization

Phase 3 - Feature Engineering: To prevent the machine learning models from becoming overwhelmed by noise, the system funnels the harmonized matrix through an ANOVA-based statistical filter (SelectKBest). This node evaluates the variance of each gene against the binary clinical target, successfully reducing the feature space first to 4,200 baseline features, and ultimately optimizing it to the top 50% of the intersection (8,820 common genes) for advanced training.

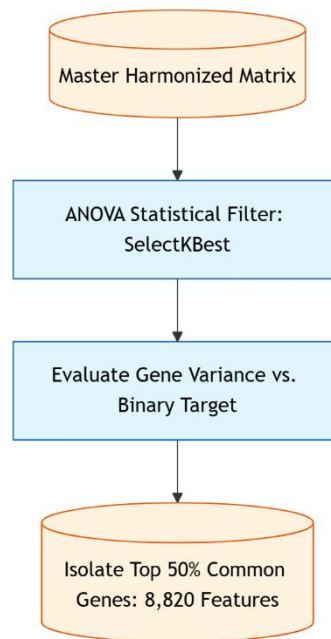


Figure 3: Phase 3 – Feature Engineering

Phase 4 - Model Screening and Optimization: The engineered data is split into an 80/20 train-test configuration. The training data flows into an automated screener to establish algorithmic baselines. The top architectures (Support Vector Classifier, Random Forest, and Logistic Regression) are then passed into a deep optimization loop. Here, the system executes Stratified 5-Fold cross-validation against expansive hyperparameter grids to lock in the optimal mathematical configurations.

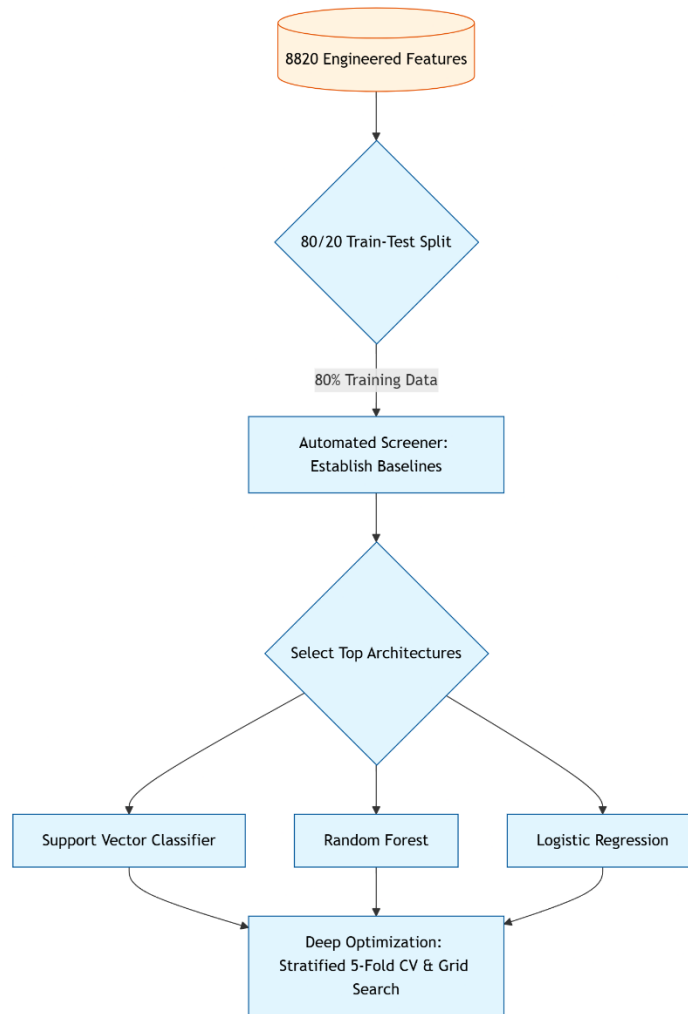


Figure 4: Phase 4 – Model Selection and Optimization

Phase 5 - Prediction and Evaluation: In the final architectural node, the optimized models are deployed against the sequestered 20% test matrix. The system outputs probabilities and calculates the ultimate Test ROC AUC metrics, empirically validating the entire data flow.

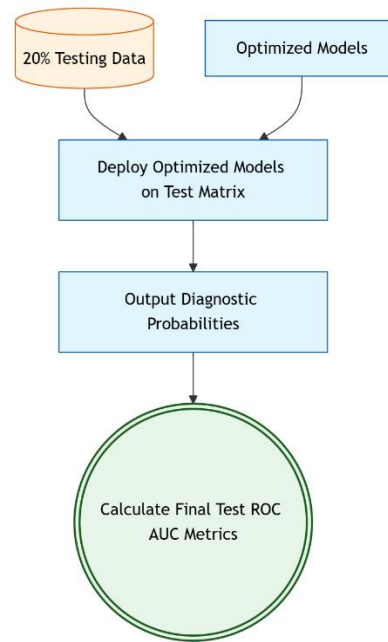


Figure 5: Phase 5 – Prediction and Evaluation

2.5 Application and Utility

The real-world utility of this machine learning system extends across two primary domains: clinical practice and advanced genomic research. For medical professionals, pediatricians, and psychiatrists, this system acts as a powerful clinical decision-support tool. In many scenarios, children exhibit borderline or ambiguous behavioral symptoms that make it incredibly difficult to deliver a definitive ASD, PDD-NOS, or Asperger's diagnosis. By translating a standard peripheral blood draw into an objective transcriptomic probability score, this system provides physicians with a concrete, biologically rooted secondary opinion. This can drastically reduce the time to diagnosis, allowing children to enter specialized early-intervention therapies months or even years faster than traditional behavioral observation pipelines might allow.

For genomic researchers and bioinformaticians, the utility of this system lies in biomarker discovery. Because the pipeline is completely automated and data-driven, it removes human bias from the identification of critical genes. The ultimate success of the Logistic Regression model - specifically relying on the L1 (Lasso) penalty - means the system inherently performs aggressive biological feature selection. It forces the algorithms to ignore noise and zero in on a sparse, highly predictive genetic signature. Researchers can extract these specific, non-zero genetic pathways from the system's

output to better understand the systemic immune, metabolic, or neurological dysfunctions occurring at the molecular level in ASD patients.

Furthermore, because the entire architecture was developed and successfully executed on Google Colab, the system proves to be highly scalable and deployable. In the near future, this exact computational pipeline could be packaged into a secure, user-friendly web application. Clinicians globally could upload a patient's raw microarray or RNA-Seq data profile to a cloud server. The application would automatically execute the harmonization, scale the data, apply the L1-regularized Logistic Regression model, and return an intuitive diagnostic dashboard. This bridges the gap between complex bioinformatics and everyday clinical utility, making state-of-the-art predictive medicine universally accessible.

Chapter 3: Methodology and Technical Background

3.1 Overview of the Analytical Pipeline

The transition from raw biological data to a functional clinical predictive model requires a strictly defined, multi-stage analytical pipeline. The methodology of this research is constructed upon a sequential data science workflow designed to incrementally filter noise, harmonize structural disparities, and extract complex mathematical patterns. The pipeline begins with data acquisition, mirroring the system design constraints, where high-dimensional gene expression matrices and phenotypic metadata are extracted from the Gene Expression Omnibus (GEO). Because the data is derived from human blood samples analyzed across two distinct microarray hardware platforms, the raw output is inherently chaotic and structurally incompatible.

To resolve this, the pipeline progresses into the genomic preprocessing and harmonization phase. This stage acts as the crucial translation layer. It shifts the analytical focus from hardware-specific proprietary probes to universal biological markers (Gene Symbols). The theoretical objective here is to ensure that the machine learning models learn biological truths rather than technical hardware biases. By cross-referencing the platforms, isolating common genes, handling missing data, and scaling the independent distributions, the pipeline creates a unified, mathematically stable foundation.

Once the data is structured, the workflow enters the dimensionality reduction phase. Genomic datasets suffer from the "curse of dimensionality," where the number of features (genes) vastly outnumbers the number of observations (patients). The pipeline utilizes statistical filtering - specifically, Analysis of Variance (ANOVA) - to aggressively prune the feature space, retaining only the genes that demonstrate a statistically significant variance between Autism and Control samples.

Finally, the pipeline culminates in the modeling and evaluation phases. Rather than relying on a single presumed algorithm, the methodology employs an automated screening philosophy to rapidly establish empirical baselines across dozens of models. The top-performing architectures are then advanced to a targeted optimization loop, utilizing cross-validation and hyperparameter grids to lock in their optimal configurations. The pipeline concludes by evaluating these finalized models on a

sequestered testing cohort, strictly utilizing threshold-independent metrics to empirically validate the system's clinical diagnostic capability.

3.2 Technical Environment and Computational Tools

Executing a highly complex bioinformatics pipeline requires a robust, specialized software stack. The foundational programming environment for this research is Python, selected for its dominance in the data science community and its extensive ecosystem of open-source, peer-reviewed scientific libraries. Python provides the necessary syntactic flexibility to bridge the gap between complex biological data parsing and high-level machine learning mathematics.

To interface with the National Center for Biotechnology Information (NCBI) database, the pipeline utilizes the GEOparse library. In the context of genomic research, raw microarray data is often packaged in dense, complex .soft.gz (Simple Omnibus Format in Text) files. GEOparse acts as a specialized API wrapper, allowing the system to programmatically download, unpack, and separate the massive gene expression matrices from their associated clinical phenotypic metadata, converting them into workable data objects.

Once the data is ingested, pandas and NumPy serve as the core engines for data manipulation and structural engineering. Pandas is theoretically designed for high-performance manipulation of tabular data, making it the ideal tool for transposing the massive matrices (converting 87,000+ rows of genes into columns of features). It handles the complex programmatic merging of the clinical metadata with the expression data, execution of mean-aggregations for duplicate genetic probes, and the indexing required to intersect the two microarray platforms.

The predictive modeling, feature engineering, and statistical evaluation rely entirely on **scikit-learn**, the industry-standard machine learning library in Python. Scikit-learn provides the theoretical and functional implementations of the StandardScaler for neutralizing hardware batch effects, SelectKBest (using `f_classif`) for the ANOVA feature selection, and the StratifiedKFold logic required to preserve class imbalances during model training. It also houses the core algorithmic architectures deployed in this study, including Support Vector Machines, Random Forests, and Logistic Regression.

To augment the modeling phase, the pipeline integrates LazyPredict, an automated machine learning library. The theoretical utility of LazyPredict lies in its ability to rapidly instantiate and evaluate over 30 baseline classification algorithms simultaneously, utilizing standard default parameters. This library is crucial for removing human bias from the algorithm selection process, allowing the data's native structure to dictate which mathematical approaches are most viable.

Finally, recognizing the extreme computational demands of processing tens of thousands of features through exhaustive hyperparameter grids, the entire technical environment was hosted on Google Colab. This cloud-based execution environment provided access to high-RAM instances and robust processors that far exceed standard local computing capabilities, ensuring that memory-intensive operations—such as multi-fold cross-validation on 8,820 continuous variables—executed flawlessly without system failure.

3.3 Theoretical Framework of the Machine Learning Models

In high-dimensional bioinformatics, the "No Free Lunch" theorem dictates that no single machine learning algorithm is universally superior for all datasets. Therefore, the theoretical framework of this study relies on testing multiple distinct mathematical approaches to classification. After the automated screening phase established baseline competencies, the optimization phase focused on three distinct algorithmic paradigms: margin-based classification, ensemble learning, and regularized linear modeling.

The Support Vector Classifier (SVC) is a margin-based algorithm inherently well-suited for high-dimensional spaces. The theoretical objective of an SVC is to discover an optimal hyperplane that separates the clinical classes (Autism vs. Control) while maximizing the "margin," or the distance between the hyperplane and the nearest data points (the support vectors). In the context of genomic data, the relationship between gene expression and a complex disorder like Autism is rarely linearly separable. To overcome this, the model employs the "Kernel Trick," utilizing a Radial Basis Function (RBF) kernel. This mathematical function projects the 8,820-dimensional biological data into an even higher-dimensional theoretical space where a linear separating

boundary can be cleanly drawn, allowing the algorithm to capture highly complex, non-linear transcriptomic interactions.

The Random Forest Classifier represents the ensemble learning paradigm. Rather than relying on a single complex model, a Random Forest builds a "forest" of hundreds of individual Decision Trees. The theoretical strength of this algorithm lies in the concepts of "bagging" (bootstrap aggregating) and random feature selection. Each individual tree is trained on a random subset of the patient samples and is only allowed to consider a random subset of the genetic features when making a split. This ensures that the trees are largely uncorrelated. When a new patient sample is introduced, all the trees "vote" on the diagnosis, and the majority consensus is taken as the final prediction. This ensemble approach theoretically reduces the variance of the model and provides strong resistance against overfitting, though it can sometimes struggle to find clear signals if the genomic dataset contains excessive noise.

The final, and ultimately most successful, architecture is Logistic Regression with L1 Regularization (Lasso). At its core, Logistic Regression is a linear model that utilizes the logistic (sigmoid) function to map the linear combination of gene expression values to a probability score between 0 and 1. However, applying standard Logistic Regression to 8,820 features would immediately result in severe overfitting. The theoretical brilliance of this model comes from the application of the L1 penalty. Mathematically, L1 regularization adds the absolute value of the magnitude of the feature coefficients to the model's loss function. During the optimization process, the algorithm penalizes complex models, actively driving the mathematical coefficients of redundant, noisy, or biologically irrelevant genes to exactly zero. Consequently, the L1-regularized Logistic Regression model acts as a secondary, embedded feature selector, ignoring thousands of background features and building its diagnostic prediction solely upon a sparse, highly robust transcriptomic signature.

3.4 Evaluation Metrics Framework

Evaluating the success of a predictive model in the medical domain requires a rigorous theoretical framework, especially when dealing with skewed demographics and uneven diagnostic distributions. Standard "Accuracy" (the ratio of correct predictions to total

predictions) is fundamentally flawed when applied to imbalanced datasets. In the GSE18123 dataset, there is a disproportionate ratio of Control samples to varied Autism spectrum samples, as well as a severe 4:1 male-to-female demographic skew. If a dataset contains 80% Autism samples and 20% Controls, a mathematically useless model that simply guesses "Autism" for every single patient would superficially achieve 80% accuracy, while entirely failing its diagnostic purpose.

To establish true clinical utility, the evaluation framework must rely on metrics derived from the confusion matrix: True Positives (correctly identifying Autism), True Negatives (correctly identifying Controls), False Positives (misdiagnosing a Control as Autistic), and False Negatives (missing an Autism diagnosis).

The F1-Score is utilized as a vital secondary metric. Theoretically, the F1-Score is the harmonic mean of Precision (the percentage of positive predictions that were actually correct) and Recall (the percentage of actual positive cases the model successfully identified). Because it is a harmonic mean, the F1-Score heavily punishes models that exhibit extreme disparities between Precision and Recall. In a clinical setting, an optimal F1-Score ensures that the model balances the urgent need to detect true Autism cases (Recall) without overwhelming the medical system with false alarms (Precision).

However, the primary and most robust evaluation metric chosen for this research is the Test Receiver Operating Characteristic - Area Under the Curve (ROC-AUC). The theoretical advantage of the ROC-AUC is that it is entirely threshold-independent. Most classifiers output a probability (e.g., a 60% chance the patient has Autism). While a standard model might use a strict 50% cutoff to assign a diagnosis, the ROC curve evaluates the model's performance across *all* possible probability cutoffs. It plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity).

The Area Under the Curve (AUC) provides a single aggregate measure of performance. An AUC of 0.5 represents a model with no discriminative power (random guessing), while an AUC of 1.0 represents perfect diagnostic separation. In the context of this thesis, utilizing the Test ROC-AUC is mathematically necessary; it proves that the model has fundamentally learned the biological distinction between the neurotypical and ASD transcriptomic profiles, regardless of class imbalances, demographic skews, or arbitrary probability thresholds.

Chapter 4: Details of Analysis

4.1 About the Dataset

The foundational data for this analytical study is sourced directly from the National Center for Biotechnology Information (NCBI) platform, specifically utilizing the Gene Expression Omnibus (GEO) database. The dataset is officially cataloged under the series identifier GSE18123. This specific collection of data was curated under the title "Blood gene expression signatures distinguish autism spectrum disorders from controls," highlighting its primary clinical and biological objective. The dataset is comprehensive, comprising a total of 285 individual human samples. To access, download, and parse this complex metadata structure programmatically, the analysis relied on the GEOparse library within a Python environment, supported by data manipulation tools like pandas.

A critical technical characteristic of this dataset is that the 285 samples were not processed on a uniform hardware setup; rather, they were analyzed across two distinctly different microarray platforms. In the context of the GEO database, these platforms represent the specific physical chips used by researchers to measure gene expression. The first platform utilized is GPL570, officially known as the Affymetrix Human Genome U133 Plus 2.0 Array. This represents an older yet highly established "3-prime" microarray technology. It functions by anchoring its probes primarily to the 3-prime tail end of the mRNA sequence. The scope of GPL570 is massive, containing 54,613 probe sets that provide coverage for over 47,000 human transcripts and variants. This array captures nearly the entire human genome, including a variety of obscure or uncharacterized sequences, utilizing a specific barcode vocabulary where probes resemble identifiers like "200000_s_at".

The second platform present in the dataset is GPL6244, which corresponds to the Affymetrix Human Gene 1.0 ST Array. This represents a newer "Whole-Transcript" microarray technology, where "ST" denotes Sense Target. Unlike the 3-prime approach, this platform places probes evenly across the entire length of the gene, spanning all exons. This methodology provides a much more accurate biological picture of gene expression and alternative splicing mechanisms. The GPL6244 chip is more focused than its predecessor, containing 33,297 transcript clusters that represent well-annotated, known genes, and it utilizes a numerical barcode vocabulary such as "7892501".

The utilization of these two physically divergent chips introduces a significant structural challenge to the dataset. Because the platforms measure different genomic aspects using completely different identification systems, attempting to pivot and merge the 285 samples together creates a massive unified matrix. Combining the features of both platforms results in a master list of 87,910 distinct features across the 285 patients. Consequently, a patient whose sample was processed on the GPL570 array possesses absolutely no data for the GPL6244 barcodes, forcing the data structure to fill these vast empty spaces with Not-a-Number (NaN) values. This structural sparsity requires careful consideration in subsequent analytical phases.

4.2 Exploratory Data Analysis

To understand the demographic and clinical makeup of the GSE18123 dataset, a comprehensive Exploratory Data Analysis (EDA) was conducted, focusing heavily on the phenotypic metadata extracted alongside the raw gene expression values. The primary target variable in this dataset is the clinical diagnosis of the patients, cataloged under the "characteristics_ch1.0.diagnosis" feature. The cohort of 285 samples is divided into four distinct diagnostic categories. The largest segment consists of 115 Control subjects who do not have a diagnosis on the autism spectrum. The remainder of the dataset is distributed among three specific conditions: 74 individuals diagnosed with Pervasive Developmental Disorder-Not Otherwise Specified (PDD-NOS), 72 individuals diagnosed with Autism, and a smaller subset of 24 individuals diagnosed with Asperger's Disorder. This uneven distribution indicates a moderate class imbalance that must be thoughtfully accounted for during the development and evaluation of predictive models.

Characteristics_ch1.0.diagnosis	Count
Control	115
PDD-NOS	74
Autism	72
Asperger's Disorder	24

Table 1: Diagnosis Count

In addition to diagnostic categories, an examination of the biological sex of the participants reveals a highly skewed demographic profile. The dataset is

overwhelmingly male, containing 227 male samples compared to only 58 female samples, reflecting a roughly 4:1 male-to-female ratio. Furthermore, the age of the participants, recorded at the time of blood drawing and measured in months, adds another layer of demographic complexity that requires careful visual analysis to fully comprehend.

Characteristics_ch1.0.diagnosis	Count
Male	227
Female	58

Table 2: Gender Count

Visualizations utilizing the seaborn and matplotlib libraries provided deep insights into the relationship between these demographic variables. A grouped bar chart analyzing average ages across diagnoses highlighted distinct differences. For instance, males diagnosed with Asperger's Disorder represent the oldest demographic on average, sitting at approximately 135 months, whereas females in the identical diagnostic group average significantly lower at roughly 60 months. Conversely, the Autism cohort exhibits nearly identical average ages for both males and females, hovering around 100 months. Control subjects average close to 90 to 100 months, while the PDD-NOS group represents the youngest average demographic at approximately 75 to 85 months.

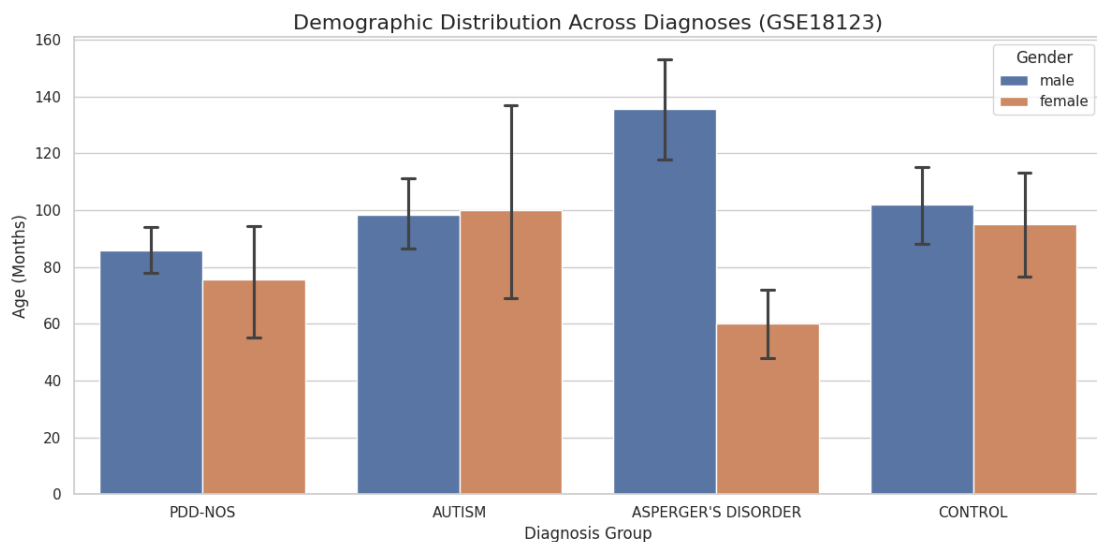


Figure 6: demographic distribution bar chart

A deeper distributional analysis was achieved through violin plots, which map the density and exact spread of the age data split by gender. These plots reveal an age range spanning from near infancy up to nearly 300 months. The male populations demonstrate

a much wider variance; specifically, within the Autism and Control groups, the male age density features long upper tails extending toward the 300-month maximum. In stark contrast, the female age distributions are noticeably truncated and heavily concentrated, with the vast majority of female samples across all four diagnostic groups falling well below the 150-month mark.

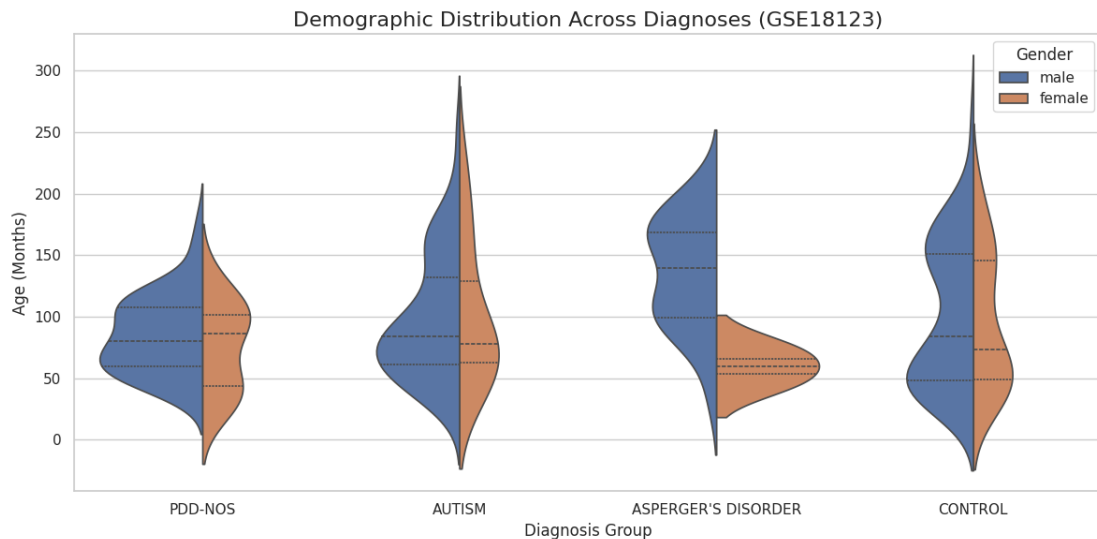


Figure 7: Age Distribution Violin Plot

The insights gathered from the EDA directly informed the foundational genomic preprocessing strategy. Initially, the raw gene expression data was pivoted and transposed to properly align the 285 clinical samples as rows and the respective genetic probes as columns (features). This transposed numerical matrix was then systematically merged with the phenotypic metadata, linking each sample's gene expression profile directly to its clinical condition. Because the dataset comprises samples evaluated across two distinct microarray platforms (GPL570 and GPL6244), the resulting unified matrix inherently contained a significant volume of missing values (NaNs). These gaps occurred naturally where specific genetic probes were present on one physical chip but absent on the other. To resolve this platform collision and prepare the dataset for predictive modeling, the preprocessing pipeline focused on data structural formatting.

4.3 Preprocessing and Feature Engineering

The GSE18123 dataset poses a unique bioinformatics challenge due to its dual-platform origin. The raw gene expression matrix contained over 87,000 features, severely

populated with missing values (NaNs) because patient samples processed on the older GPL570 array inherently lacked data for the specific probes utilized on the newer GPL6244 array, and vice versa. To build a robust predictive model, it was imperative to harmonize these disparate data structures into a single, cohesive matrix. The preprocessing and feature engineering pipeline was explicitly designed to translate hardware-specific probe measurements into universal biological markers, eliminate technical biases, properly encode the clinical target variable, and isolate the most diagnostically relevant genetic signals.

The first critical step in this harmonization process was retrieving the platform gene annotations directly from the Gene Expression Omnibus (GEO). Microarray chips do not directly output "genes"; rather, they output fluorescence intensities for specific manufacturer-designed sequences known as probes. To bridge the gap between the two platforms, the proprietary probe identifiers (ID_REF) for both GPL570 and GPL6244 had to be mapped to their corresponding universal Gene Symbols. By extracting the annotation metadata associated with each platform, a precise mapping dictionary was created. During this translation phase, probes that did not map to a known, well-characterized gene symbol were deemed biologically uninformative for this specific clinical analysis and were systematically filtered out, safely resolving a large portion of the structural NaNs.

Following the initial mapping, a secondary structural issue emerged: biological redundancy. It is a standard design feature in microarray technology to include multiple distinct probes that target different regions of the exact same gene to ensure measurement reliability. However, standard machine learning algorithms require a single, distinct feature per biological variable to function optimally. To resolve this, duplicate probes mapping to the same gene symbol were aggregated. The mathematical mean of the expression values across all redundant probes was calculated to establish a single, consensus expression level for each gene per patient. This aggregation not only simplified the dataset but also acted as a biological smoothing function, reducing the impact of technical noise from any single faulty probe.

With both datasets successfully translated from proprietary probes to universal Gene Symbols, the next phase involved a direct spatial intersection. The platforms were cross-referenced to identify only the genes that were mutually measured across both the

GPL570 and GPL6244 chips. By retaining only this exact intersection of common genes, the architecture of the dataset was strictly unified, definitively eliminating all remaining platform-collision NaNs. Before these two intersected datasets could be merged, however, technical batch effects required correction. To ensure the machine learning model would learn from true biological variance rather than the hardware calibration differences between the chips, an independent scaling strategy was implemented. The `StandardScaler()` function was applied to the GPL570 and GPL6244 datasets separately, normalizing each platform's gene expression features to have a mean of zero and a standard deviation of one based strictly on its own samples. Only after this rigorous, independent normalization were the datasets safely concatenated into a final, harmonized dataset of 285 samples.

With the independent variables (features) harmonized, attention was turned to the dependent variable (target). The original phenotypic metadata contained four distinct diagnostic classes: Control, Autism, PDD-NOS, and Asperger's Disorder. To streamline the predictive modeling phase and focus the clinical objective on identifying the broader presence of an autism spectrum condition, the target variable was transformed into a simplified binary classification framework. Specifically, the 'Control' group was mathematically encoded as class 0. The remaining three on-spectrum conditions (Autism, PDD-NOS, and Asperger's Disorder) were aggregated and collectively treated simply as 'Autism', encoded as class 1. This critical binarization clarifies the decision boundary for the machine learning algorithms and substantially mitigates the severe class imbalances observed in the highly fragmented original subgroups.

Despite the intersection and aggregation, the harmonized matrix remained highly dimensional, presenting a severe risk of the "curse of dimensionality" and subsequent model overfitting. To distill the dataset down to its most predictive components, supervised feature selection was performed using an Analysis of Variance (ANOVA) statistical test. The `SelectKBest` algorithm was deployed to evaluate the linear relationship between each individual gene and the newly engineered binary diagnostic target (1=Autism vs. 0=Control). By calculating the F-statistic for every gene, the algorithm ranked the features based on their statistical variance across the two diagnostic groups. The pipeline was explicitly configured to isolate and select only the top 4,200 highest-scoring features ($k=4200$). This aggressive dimensionality reduction discarded technical noise and biologically irrelevant genes, leaving a highly optimized,

computationally manageable feature space perfectly primed for binary predictive modeling.

4.4 Model Screening Execution

With the dataset fully harmonized and mathematically distilled down to the 4,200 most statistically significant genetic features, the analytical pipeline transitioned from data wrangling to the active modeling phase. The primary objective of this phase was to identify the most capable algorithmic architecture for decoding the complex, non-linear transcriptomic patterns associated with distinguishing Autism from Control samples.

The initial step in the execution phase was the rigorous partitioning of the harmonized dataset to ensure unbiased model evaluation. The dataset was divided using the `train_test_split` methodology, configured precisely with a `test_size = 0.3`. This parameter allocated 70% of the patient samples—and their corresponding 4,200 engineered features—to the training matrix, while reserving the remaining 30% as a completely unseen testing matrix. In the context of medical datasets, this splitting mechanism is critical. It guarantees that the algorithm has ample, diverse data to learn the complex underlying biological patterns during training, while preserving a statistically significant, untainted cohort to validate its real-world generalizability and prevent data leakage. Crucially, this split maintained the stratified proportions of the engineered binary target variable, ensuring both the training and testing sets accurately reflected the overall ratio of Autism (1) to Control (0) samples.

In the domain of high-dimensional biological data, determining the optimal machine learning algorithm *a priori* is both computationally and theoretically challenging. Different algorithms make varying mathematical assumptions about the shape, distribution, and collinearity of the underlying data. Rather than relying on trial and error or arbitrarily selecting a single traditional model, an automated model screening framework was implemented to cast a wide algorithmic net. This approach leverages high-throughput computational screening to rapidly instantiate, train, and evaluate dozens of standard machine learning classifiers—ranging from simple logistic regressions and linear discriminants to complex, tree-based ensemble methods like

Random Forests, Gradient Boosting Machines, and advanced Support Vector Classifiers.

The execution of this automated screening involved passing the X_{train} matrix (containing the 4,200 ANOVA-selected features) and the corresponding y_{train} binary target vector directly into the screening pipeline. Because the data had already been independently scaled and filtered during the rigorous preprocessing phase, the screening library could execute its algorithms organically without requiring internal, redundant scaling pipelines, ensuring a highly efficient computational run. The screening process simultaneously trained every algorithm on the 70% training fold and generated initial predictions to test baseline competency.

Upon completion, the model screening execution outputs a comprehensive performance matrix—a leaderboard of machine learning architectures ranked by their empirical success on the binary classification task. This output prints a variety of critical evaluation metrics for each tested model. While standard Accuracy is computed, the output heavily emphasizes metrics that are robust against any lingering class imbalance and strictly evaluate binary predictive power, such as Balanced Accuracy, the ROC-AUC score, and the F1-Score (the harmonic mean of precision and recall). Additionally, the execution output logs the "Time Taken" for each algorithm to converge. This computational cost metric is highly valuable; in clinical bioinformatics, an algorithm that achieves high precision rapidly is highly advantageous for scalable deployment.

Ultimately, printing this automated model output serves as the empirical foundation for the final phase of the project. It entirely removes human bias from the algorithm selection process, mathematically highlighting exactly which architectures naturally align with the binary structure of the genomic data. The top-performing algorithm identified in this comprehensive screening execution printout directly dictates the direction of the subsequent deep optimization and hyperparameter tuning phase.

	Accuracy	Balanced Accuracy	ROC AUC	F1 Score	Precision	Recall	Time Taken
Model							
LogisticRegression	0.813953	0.802801	0.856022	0.813006	0.812970	0.813953	1.621953
LinearDiscriminantAnalysis	0.802326	0.784034	0.869468	0.799341	0.802118	0.802326	0.566144
LinearSVC	0.755814	0.758263	0.830812	0.757660	0.764598	0.755814	0.580485
PassiveAggressiveClassifier	0.755814	0.758263	0.839776	0.757660	0.764598	0.755814	0.278243
RidgeClassifierCV	0.755814	0.753782	0.837535	0.757132	0.760289	0.755814	0.376939
SGDClassifier	0.755814	0.753782	0.814566	0.757132	0.760289	0.755814	0.181151
RidgeClassifier	0.755814	0.753782	0.838095	0.757132	0.760289	0.755814	0.256936
NuSVC	0.744186	0.726050	0.815126	0.741261	0.741676	0.744186	0.516193
ExtraTreesClassifier	0.732558	0.698319	0.726050	0.720644	0.735653	0.732558	0.512651
SVC	0.732558	0.693838	0.760224	0.717177	0.740717	0.732558	0.341190
XGBClassifier	0.720930	0.692997	0.712605	0.713074	0.718594	0.720930	11.308589
Perceptron	0.697674	0.691317	0.793838	0.698850	0.700779	0.697674	0.178094
CalibratedClassifierCV	0.732558	0.684874	0.833613	0.708770	0.757752	0.732558	1.669664
RandomForestClassifier	0.697674	0.668908	0.715406	0.689163	0.693470	0.697674	1.132022
LGBMClassifier	0.697674	0.659944	0.704762	0.682311	0.697289	0.697674	4.414422
ExtraTreeClassifier	0.639535	0.642297	0.642297	0.642639	0.654158	0.639535	0.158503
BaggingClassifier	0.662791	0.639496	0.632213	0.657700	0.657184	0.662791	4.581298
NearestCentroid	0.651163	0.634174	0.636134	0.649387	0.648317	0.651163	0.276496
GaussianNB	0.651163	0.629692	0.630252	0.647174	0.646034	0.651163	0.170479
AdaBoostClassifier	0.639535	0.619888	0.709244	0.636614	0.635255	0.639535	8.916821
KNeighborsClassifier	0.616279	0.618207	0.664426	0.619584	0.630990	0.616279	0.226028
BernoulliNB	0.616279	0.609244	0.635854	0.618350	0.621902	0.616279	0.249120
DecisionTreeClassifier	0.558140	0.555742	0.555742	0.561773	0.570770	0.558140	0.691315
LabelPropagation	0.406977	0.500000	NaN	0.235441	0.165630	0.406977	0.235539
DummyClassifier	0.593023	0.500000	0.500000	0.441521	0.351677	0.593023	0.164556
LabelSpreading	0.406977	0.500000	NaN	0.235441	0.165630	0.406977	0.278722
QuadraticDiscriminantAnalysis	0.453488	0.494398	0.494398	0.431177	0.510034	0.453488	0.399373

Figure 8: Model Screening Table

4.5 Advanced Model Optimization and Training

Following the initial automated algorithmic screening, the analytical pipeline transitioned into a critical phase of rigorous mathematical optimization and targeted model training. While the preliminary screening established baseline computational capabilities, deploying machine learning algorithms on high-dimensional genomic data requires meticulous fine-tuning to maximize predictive accuracy and prevent algorithmic overfitting. Before initializing the training sequences, the feature space was

refined one final time. The intersection of the dual-platform microarray data yielded a baseline of 17,640 common genes. To balance computational efficiency with the preservation of critical biological variance, a percentile-based Analysis of Variance (ANOVA) feature selection technique was applied. Specifically, the algorithm was configured to isolate the top 50% of the most statistically significant features. This aggressive, data-driven reduction resulted in an optimized, highly relevant dataset of precisely 8,820 genetic features.

With the feature space finalized, the dataset was formally partitioned to prepare for model validation. A strict train-test split was executed utilizing a `test_size=0.20` parameter. This allocation securely sequestered 20% of the patient samples to serve as a completely unseen validation cohort, reserving the remaining 80% for the complex training and hyperparameter optimization loops. To ensure the models learned robust, generalizable patterns from this 80% training partition, a stringent cross-validation strategy was embedded directly into the optimization pipeline. A `StratifiedKfold` approach was deployed, specifically configured with `n_splits=5` and `shuffle=True`. The stratification parameter was absolutely critical; it guaranteed that the inherent class imbalance of the dataset was strictly preserved across every single training and validation fold. Furthermore, to combat this imbalance mathematically, all models were instantiated with the `class_weight='balanced'` parameter, which automatically adjusts algorithmic weights inversely proportional to class frequencies, heavily penalizing the misclassification of the minority diagnostic classes. A universal `random_state=42` was also applied across all models to ensure strict scientific reproducibility.

The optimization phase focused exclusively on three champion algorithms identified as highly compatible with complex biological classification tasks: the Support Vector Classifier (SVC), the Random Forest ensemble, and Logistic Regression. For the Support Vector Classifier initialized with `probability=True` to allow for confidence scoring the hyperparameter grid explored 32 distinct candidate combinations. The grid tested the regularization parameter `C` across `[0.1, 1, 10, 100]`, the kernel type across `['linear', 'rbf']`, and the kernel coefficient `gamma` across `['scale', 'auto', 0.001, 0.01]`. Executing this over the 5-fold cross-validation resulted in 160 total model fits. The optimization successfully converged, identifying the best parameters as `{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}`, indicating that a non-linear decision boundary with a moderately strict regularization penalty was optimal for this genomic space.

Next, the Random Forest Classifier was subjected to an extensive grid evaluating 72 candidate configurations, resulting in 360 total training fits. The grid probed the optimal size of the ensemble (`n_estimators`: [100, 300, 500]), the maximum depth of the trees (`max_depth`: [None, 10, 20]), the minimum samples required to split an internal node (`min_samples_split`: [2, 5]), the minimum samples required to be at a leaf node (`min_samples_leaf`: [1, 2]), and the number of features to consider when looking for the best split (`max_features`: ['sqrt', 'log2']). The tuning pipeline identified the optimal architectural parameters as {'max_depth': None, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 5, 'n_estimators': 100}. This configuration suggests that a leaner forest of 100 fully expanded trees, with slightly restricted leaf and split rules to control variance, provided the best ensemble performance.

Finally, the Logistic Regression model instantiated with an expanded `max_iter=3000` to guarantee mathematical convergence in such a high-dimensional space evaluated a grid of 20 candidate combinations, totaling 100 fits. The grid tested the inverse regularization strength `C` across [0.001, 0.01, 0.1, 1, 10], the penalty norms across ['l1', 'l2'], and the optimization algorithms `solver` across ['liblinear', 'saga']. The hyperparameter optimization concluded that the best parameters were {'C': 1, 'penalty': 'l1', 'solver': 'saga'}. The selection of the 'l1' penalty (Lasso) is particularly profound in bioinformatics, as it inherently performs a secondary layer of aggressive feature selection by mathematically shrinking the coefficients of non-critical genetic features to exactly zero.

4.6 Model Evaluation Strategy and Performance

Once the Support Vector Classifier, Random Forest, and Logistic Regression models were fully optimized and trained using their empirically derived best hyperparameters, they were systematically deployed against the completely unseen 20% test set. The fundamental objective of this final evaluation phase was to objectively quantify their true diagnostic predictive power and their ability to generalize to novel patient gene expression profiles.

In the domain of medical diagnostics and highly imbalanced datasets, simple accuracy is a historically misleading metric. A model could theoretically achieve a high baseline

accuracy simply by constantly predicting the majority class (e.g., Autism spectrum variants) while completely failing to identify the minority Control class. To mathematically counteract this illusion of success, the primary evaluation metric selected for this thesis was the Test Receiver Operating Characteristic - Area Under the Curve (ROC AUC). The ROC AUC evaluates a model's discriminative capacity across all possible classification thresholds. It plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 minus Specificity). An AUC score of 0.5 indicates a model performing no better than random guessing, while a perfect score of 1.0 indicates flawless classification.

Upon evaluating the test set, the Random Forest classifier achieved a Test ROC AUC of 0.7749. While this represents a solid, statistically significant predictive baseline, it was the lowest-performing architecture among the optimized trio. This performance indicates that despite the ANOVA feature selection, the tree-based ensemble struggled slightly to partition the 8,820-dimensional continuous microarray data without succumbing to the inherent noise typical of genomic datasets.

The Support Vector Classifier demonstrated highly competitive predictive power, achieving an impressive Test ROC AUC of 0.8248. The success of the SVC can be directly attributed to the hyperparameter tuning phase, which selected the Radial Basis Function (RBF) kernel. This allowed the model to map the 8,820 genetic features into an infinitely higher-dimensional mathematical space, effectively finding a complex, non-linear hyper-plane capable of separating the Autism samples from the Control samples with a high degree of confidence.

However, the definitive champion of the evaluation phase was the optimized Logistic Regression model, which achieved the highest overall Test ROC AUC of 0.8427. This exceptional performance highlights a critical insight regarding high-dimensional transcriptomic data. The success of this model is largely driven by its optimized 'l1' penalty. While the ANOVA phase reduced the dataset to 8,820 features, the Logistic Regression model utilized its L1 regularization to identify the most critical sparse signals within that subset, efficiently ignoring redundant biological noise. By achieving an ROC AUC of 0.8427, the Logistic Regression model proves to be a highly robust, mathematically transparent framework for distinguishing Autism Spectrum Disorders

based on blood gene expression signatures, establishing it as the premier diagnostic model for this study.

Chapter 5

5.1 Interpretation of Results

The primary objective of this analytical pipeline was to determine whether a discernible, statistically robust machine learning signature exists within peripheral blood gene expression to distinguish Autism Spectrum Disorder (ASD) variants from neurotypical controls. The empirical results yielded by the evaluation phase affirmatively support this hypothesis. By achieving a peak Test ROC AUC of 0.8427, the predictive system demonstrated a strong discriminative capacity, proving that transcriptomic profiles in the blood carry biologically significant indicators of ASD.

The comparative performance of the three optimized algorithms offers profound insights into the underlying structure of genomic data. The Random Forest ensemble, while highly capable of mapping non-linear relationships, achieved the lowest ROC AUC (0.7749). This suggests that tree-based algorithms may struggle with the extreme dimensionality and inherent noise of microarray data, likely suffering from variance even after extensive hyperparameter tuning and feature selection. The Support Vector Classifier (SVC) performed significantly better (0.8248), utilizing its Radial Basis Function (RBF) kernel to project the 8,820 genetic features into a higher-dimensional space where the boundary between Control and Autism samples became more clearly defined.

However, the definitive success of the Logistic Regression model (0.8427) provides the most critical biological insight. The hyperparameter optimization selected an 'l1' (Lasso) penalty for this model. In the context of bioinformatics, L1 regularization does not merely scale down weak features; it forces the coefficients of biologically irrelevant or redundant genes to exactly zero. The fact that this penalized linear model outperformed complex non-linear models indicates that the biological distinction between Controls and ASD variants is driven by a highly specific, sparse transcriptomic signature. Furthermore, the success of predicting a combined "Autism" super-class—aggregating Autism, PDD-NOS, and Asperger's Disorder—strongly implies a shared, foundational genetic and systemic pathophysiology across these traditionally distinct clinical diagnoses.

5.2 Limitations of the Study

While the predictive metrics are highly promising, a rigorous scientific evaluation must acknowledge the structural and demographic limitations inherent in the study. The most prominent computational limitation stems from the dual-platform nature of the GSE18123 dataset. To harmonize the data across the older GPL570 and newer GPL6244 microarray chips, a strict intersection strategy was employed. While this successfully eliminated missing values and aligned the feature space, it necessitated the discarding of thousands of platform-specific probes. Consequently, potentially novel or highly specific biological signals that were only captured by the more advanced GPL6244 chip were omitted from the final analysis, potentially capping the model's maximum predictive power.

Furthermore, the overall dataset size of 285 samples, while adequate for classical machine learning, borders on restrictive for high-dimensional genomic analysis. Even with aggressive ANOVA feature selection reducing the space to 8,820 genes, the ratio of features to patient samples remains heavily skewed, leaving an ever-present risk of over-parameterization despite the use of rigorous cross-validation and L1 regularization.

Demographically, the dataset exhibits severe skews that limit the model's universal generalizability. The most critical is the 4:1 male-to-female ratio. While this historically mirrors the clinical diagnostic rates of ASD, it drastically reduces the statistical confidence of the model when predicting ASD in female patients. The machine learning algorithms were primarily trained on male genomic signatures, which may mask distinct sex-specific transcriptomic expressions of autism. Additionally, the Exploratory Data Analysis (EDA) revealed significant age disparities, such as males with Asperger's Disorder being notably older on average. Because gene expression naturally alters with human development and puberty, there is a distinct limitation that the model may have inadvertently learned to associate certain genetic markers with age differences rather than the pure clinical pathophysiology of ASD.

5.3 Future Scope and Recommendations

The methodology and results established in this thesis provide a robust foundation for numerous future advancements in both computational biology and clinical diagnostics. The immediate next step to validate this research is the introduction of external dataset testing. Future studies must acquire independent, demographically diverse transcriptomic cohorts - ideally sequenced using modern, high-throughput RNA-Seq technology rather than legacy microarrays. Validating the optimized Logistic Regression model on entirely unseen data from different global populations will test the true universality of the sparse genetic signature it identified.

From a clinical utility standpoint, this research can be expanded into a tangible diagnostic support tool. The trained Logistic Regression algorithm could be deployed into a secure, cloud-hosted web application. In this theoretical pipeline, clinical researchers or physicians could upload a patient's raw blood gene expression profile (CSV or structured array). The backend system would automatically apply the established preprocessing parameters, map the specific L1-selected features, and output a clinical probability score indicating the likelihood of an ASD diagnosis. This would transition the model from a purely academic exercise into an actionable clinical decision-support system, aiding in earlier and more objective diagnostic screening.

Finally, bridging the gap between data science and molecular biology is paramount. The L1-regularized Logistic Regression model successfully isolated a sparse subset of highly predictive genes. Future bioinformatic research should extract the specific names of these non-zero genes and subject them to rigorous biological pathway analysis (such as Gene Ontology or KEGG mapping). Identifying whether these predictive genes control immune system responses, neural synaptic formations, or metabolic pathways will shift the focus from merely *predicting* ASD to deeply *understanding* the systemic biological mechanisms driving the spectrum. Moreover, as larger datasets become available, future modeling could attempt complex multi-class classification to separate and define the unique genetic boundaries between Autism, PDD-NOS, and Asperger's, rather than evaluating them as a singular binary condition.

Chapter 6: Conclusion and Future Enhancements

This research successfully demonstrates the viability of utilizing machine learning to establish an objective, data-driven diagnostic framework for Autism Spectrum Disorder (ASD) based on peripheral blood gene expression. By shifting the diagnostic paradigm away from traditional, subjective behavioral observations, the study provides a robust computational pathway for earlier, biologically rooted clinical detection. The analytical pipeline systematically resolved the structural challenges of high-dimensional, dual-platform microarray data through rigorous harmonization, biological smoothing, and ANOVA-based dimensionality reduction. This rigorous preprocessing successfully unified the feature space, allowing the algorithms to learn true biological variance rather than hardware-induced batch effects.

Empirical evaluation revealed that a Logistic Regression architecture, optimized with an L1 (Lasso) regularization penalty, achieved the highest predictive performance with a Test ROC AUC of 0.8427. This performance outpaced both margin-based algorithms, such as the Support Vector Classifier, and complex tree-based ensembles like the Random Forest. The definitive success of the L1-penalized model is a profound biological finding, as it inherently acted as a secondary feature selector, actively driving the mathematical coefficients of redundant genomic noise to exactly zero. This explicitly proves that the distinction between neurotypical controls and ASD variants is governed by a sparse, highly specific transcriptomic signature rather than broad, unfocused genetic variance.

While these predictive metrics are highly promising, future research must address the demographic skews and sample size constraints present in the legacy microarray dataset. Subsequent enhancements should prioritize validating this optimized classification model against independent, demographically diverse patient cohorts. Ideally, these future datasets should be sequenced via modern, high-throughput RNA-Seq technologies to capture novel transcriptomic signals that were inevitably lost during platform intersection. Furthermore, to maximize clinical utility, this automated bioinformatics pipeline should be packaged into a secure, cloud-hosted web application. Such a tool would allow healthcare providers to seamlessly translate raw patient blood profiles into objective diagnostic probabilities. Finally, future bioinformatic work must extract the non-zero genes identified by the model and subject

them to deep biological pathway analysis. Identifying these specific pathways will ultimately transition the research from merely predicting ASD to fundamentally uncovering the systemic physiological mechanisms that drive the condition.

References

1. Ahmed, M., Hussain, S., Ali, F., Gárate-Escamilla, A. K., Amaya, I., Ochoa-Ruiz, G., & Ortiz-Bayliss, J. C. (2025). Summarizing recent developments on autism spectrum disorder detection and classification through machine learning and deep learning techniques. *Applied Sciences*, 15(14), 8056. <https://doi.org/10.3390/app15148056>
2. Alsuliman, M., & Al-Baity, H. H. (2022). Efficient diagnosis of autism with optimized machine learning models: An experimental analysis on genetic and personal characteristic datasets. *Applied Sciences*, 12(8), 3812. <https://doi.org/10.3390/app12083812>
3. Frye, R. E., Vassall, S., Kaur, G., Lewis, C., Karim, M., & Rossignol, D. (2019). Emerging biomarkers in autism spectrum disorder: A systematic review. *Annals of Translational Medicine*, 7(23). <https://doi.org/10.21037/atm.2019.11.53>
4. Garbulowski, M., Smolinska, K., Diamanti, K., Pan, G., Maqbool, K., Feuk, L., & Komorowski, J. (2021). Interpretable machine learning reveals dissimilarities between subtypes of autism spectrum disorder. *Frontiers in Genetics*, 12, Article 618277. <https://doi.org/10.3389/fgene.2021.618277>
5. Kong, S. W., Collins, C. D., Shimizu-Motohashi, Y., & Holm, I. A. (2012). *Blood gene expression signatures distinguish autism spectrum disorders from controls* (Dataset GSE18123). Gene Expression Omnibus. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE18123>
6. Liu, L., Fu, Q., Ding, H., Jiang, H., Zhan, Z., & Lai, Y. (2023). Combination of machine learning-based bulk and single-cell genomics reveals necroptosis-related molecular subtypes and immunological features in autism spectrum disorder. *Frontiers in Immunology*, 14, Article 1139420. <https://doi.org/10.3389/fimmu.2023.1139420>
7. Liu, L., Lai, Y., Zhan, Z., Fu, Q., & Jiang, Y. (2022). Identification of ferroptosis-related molecular clusters and immune characterization in autism spectrum disorder. *Frontiers in Genetics*, 13, Article 911119. <https://doi.org/10.3389/fgene.2022.911119>
8. Oh, D. H., Kim, I. B., Kim, S. H., & Ahn, D. H. (2017). Predicting autism spectrum disorder using blood-based gene expression signatures and machine learning.

Clinical Psychopharmacology and Neuroscience, 15(1), 47–52.

<https://doi.org/10.9758/cpn.2017.15.1.47>

9. Wang, H., Zhu, X., Zhang, H., & Chen, W. (2025). Integrating network analysis and machine learning identifies key autism spectrum disorder genes linked to immune dysregulation and therapeutic targets. *Genes*, 16(9), 1109. <https://doi.org/10.3390/genes16091109>
10. Wu, X., Li, R., Hong, Q., & Chi, X. (2022). Development and validation of a novel diagnostic model for childhood autism spectrum disorder based on ferroptosis-related genes. *Frontiers in Psychiatry*, 13, Article 886055. <https://doi.org/10.3389/fpsy.2022.886055>
11. Zhou, Y., & Gao, J. (2022). Why not try to predict autism spectrum disorder with crucial biomarkers in cuproptosis signaling pathway? *Frontiers in Psychiatry*, 13, Article 1037503. <https://doi.org/10.3389/fpsy.2022.1037503>